

Problem-Solution Fit

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The canvas below checks that the EduGenie solution addresses the root causes, behaviours and limitations of its target learners. Each box is filled in the same position as the original visual template.

1. CUSTOMER SEGMENT(S) [CS] School and college students and self-learners who want quick, simple and interactive help with studying, from beginner to advanced level.	6. CUSTOMER LIMITATIONS [CL] e.g. budget, devices Little or no budget for paid AI subscriptions, ordinary laptops or phones with modest hardware, limited time to search and sort resources, and basic skill at writing AI prompts.	5. AVAILABLE SOLUTIONS [AS] Pros & cons General AI chatbots give good answers but are not built for studying: no ready quiz or roadmap, and results depend on how well the learner writes the prompt. Search engines return scattered links. Textbooks, study portals and video courses are static and give no feedback on understanding.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Doubts and difficult topics come up daily. Practice questions are needed before every test. A clear roadmap is missing each time a new subject or skill is started. Long notes are hard to revise quickly.	9. PROBLEM ROOT / CAUSE [RC] Most study resources are static and one-size-fits-all, and the useful tools (answers, simple explanations, quizzes, summaries, roadmaps) are spread across many sites, so learners waste time collecting and sorting information.	7. BEHAVIOR + ITS INTENSITY [BE] High-intensity cramming just before exams, mixed with steady, moderate daily study. Learners jump between search tabs, videos and notes to piece answers together.

3. TRIGGERS TO ACT [TR] An upcoming test or exam, an assignment deadline, a concept that will not make sense, or the wish to learn a new skill such as SQL and needing a clear plan.	10. YOUR SOLUTION [SL] EduGenie, a free web-based learning assistant built on FastAPI. It combines the Google Gemini API with the lightweight local LaMini-Flan-T5 model to answer questions, explain concepts in simple words, generate a 3-question quiz with instant feedback, summarize long text and build a beginner-to-advanced learning path. No login is needed, it keeps no user data, and it is live online on Vercel.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Learners open the EduGenie web page in any browser (live on Vercel). Questions, quizzes, summaries and learning paths are generated by the Google Gemini API.
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Overwhelmed by heavy notes, anxious about exams and unsure of their level, and lost without a clear path. AFTER: Relieved by short, clear summaries, confident from instant quiz feedback, and motivated by a step-by-step roadmap.		OFFLINE When EduGenie is run on the learner's own computer, concept explanations come from the local LaMini-Flan-T5 model, which needs no internet connection after its one-time download. Features that use Gemini need internet.

Fit summary: the pains (2) come from static, scattered study resources (9). EduGenie (10) answers them in one place, within the limitations of learners who have little budget and ordinary devices (6), and it can be reached through the channels learners already use (8).